

1. Number Base Representation & Conversions

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Base System Foundations

A number system is defined by its base (radix) b . A numeral in base b employs digits from 0 to $(b - 1)$.

Base 10 = Decimal (Denary)

Base 2 = Binary (used in computing)

Base 8 = Octal, Base 16 = Hexadecimal.

Conversion Methods

- Any Base to Base 10: Expand each digit by multiplying with base raised to its place value.

Example: $1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 8 + 4 + 0 + 1 = 13_{10}$.

- Base 10 to Any Base: Continuous division by target base and collecting remainders in reverse order.

2. Binary Arithmetic & Modular Arithmetic

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Binary Addition Rules

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10, (0 \text{ carry } 1)$$

$$1 + 1 + 1 = 11, (1 \text{ carry } 1)$$

Modular (Clock) Arithmetic

In arithmetic modulo n , numbers wrap around after reaching n .

a "a $b \pmod{n}$ " means that $(a - b)$ is evenly divisible by n .

Example: In modulo 7 (Days of the week), $15 \pmod{7} = 1$ because $15 = 2 \times 7 + 1$.

3. Examination Questions & Solutions

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Sample Problem

Solve for x if $24_x = 18_{10}$.

Solution: $2(x^1) + 4(x^0) = 18$

$$2x + 4 = 18$$

$$2x = 14 \Rightarrow x = 7.$$

Verify: In base 7, digits 2 and 4 are valid since both are less than 7.

Quick Reference Tips

- The maximum digit in base n is always $(n - 1)$.
- Multiplicative inverse of a modulo m exists if and only if $\gcd(a, m) = 1$.